Quantum Mechanics I — Exercise Sheet 5

Compatible and incompatible observables

Variant: Questions only

Conventions, as in Lecture 5: two observables $A = \sum_a a\,E_a$ and $B = \sum_b b\,G_b$ are read at once, with no evolution between the readings, so that $\mathrm{Prob}(a \text{ then } b) = \mathrm{Tr}(\rho E_a G_b E_a)$; for a qubit, $\rho = \frac{1}{2}(\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma})$.

Exercise 1 (the order of two spin readings). A qubit with Bloch vector $\mathbf{r}$ is measured along z and then along x, or in the other order.

1. Show that $\mathrm{Prob}(z = s \text{ then } x = q) = \frac{1}{4}(1 + s r_z)$ and $\mathrm{Prob}(x = q \text{ then } z = s) = \frac{1}{4}(1 + q r_x)$, for $s, q = \pm 1$.
2. For which states do the two orders give the same four probabilities? Why does this not make σ_z and σ_x compatible?

Exercise 2 (polarisers). Use the ideal linear polarisers of the box *The physics* of Lecture 5: light polarised at the angle β to the axis passes with the fraction $\cos^2\beta$ of its intensity, and an unpolarised beam with the fraction $\frac{1}{2}$.

1. An unpolarised beam meets polarisers at the angles 0, α and $\pi/2$, in this order. Show that the fraction $\frac{1}{8}\sin^2 2\alpha$ of the light comes through, and find the largest fraction.
2. Between the polarisers at 0 and $\pi/2$, put $N-1$ polarisers at the angles $k\pi/(2N)$, $k = 1, \ldots, N-1$. Find the fraction that comes through, its values for $N = 1$, 2 and 10, and its limit as $N \to \infty$.

Exercise 3 (a complete set on $\mathbb{C}^4$). On $\mathbb{C}^4$ with the orthonormal basis $e_1, \ldots, e_4$, let $A = \mathrm{diag}(1, 1, 2, 2)$ and $B = \mathrm{diag}(0, 1, 0, 1)$.

1. Show that A and B form a complete set, although neither does alone.
2. Find one observable whose single reading gives the joint reading of A and B.
3. Let $B' = |e_1\rangle\langle e_1| + |w\rangle\langle w|$ with $w = \frac{1}{\sqrt{2}}(e_3 + e_4)$. Show that A and B' form a complete set too, and that B and B' are not compatible.

Exercise 4 (Schrödinger's form of the uncertainty relation). Let A, B be observables, ρ a density matrix, and A_0, B_0, z as in the proof of Theorem 5.7. Write $\{X, Y\} = XY + YX$.

1. Keep the real part of z in that proof, and show that
$$(\Delta A)^2 (\Delta B)^2 \geq \tfrac{1}{4}\,|\mathrm{Tr}(\rho\,[A, B])|^2 + \tfrac{1}{4}\big(\mathrm{Tr}(\rho\{A_0, B_0\})\big)^2 .$$
2. For a qubit, take $A = \sigma_x$ and $B = \sigma_y$. Find the right side in terms of the Bloch vector, and show that the inequality is an equality exactly for the pure states.
3. For the pure state with $\mathbf{r} = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0)$, compare the two bounds with $(\Delta\sigma_x)^2(\Delta\sigma_y)^2$.

Exercise 5 (⋆ the unpolarised state hides the order). Let $\rho = \frac{1}{n}\mathbb{1}$ on $\mathbb{C}^n$. Show that for any two observables A and B, compatible or not, the two orders of reading give the same joint distribution in the state ρ. Check it for σ_z and σ_x.